

(3)

**B(6th Sm.)-Computer Sc.-G/
SEC-B-1 & SEC-B-2/CBCS**

**SEC-B-2
(Information Security)**

Full Marks : 80

Group - A

1. Answer *any ten* questions :

2×10

- (a) How is Encryption different from Hashing?
- (b) How does 'denial of service' prevent the management of communication facilities?
- (c) Explain in brief about Caesar's Cipher with suitable example.
- (d) Define Rail fence technique with a suitable example.
- (e) Distinguish between stream and block cipher.
- (f) Highlight the main differences between active and passive attack.
- (g) What is firewall?
- (h) What is Encapsulating Security Payload?
- (i) What is Ring?
- (j) Find $2^{15} \bmod 5$ using Fermat's theorem.
- (k) Write the main differences between worms and viruses.
- (l) What is Trojan horse?
- (m) What do you understand by Brute-force-attack?

Group - B

2. Answer *any four* questions :

5×4

- (a) Explain Euler's theorem with suitable example.
- (b) What is IP Sniffing and Spoofing?
- (c) Write short note on Steganography.
- (d) Discuss briefly about SSL attacks.
- (e) Briefly describe how the security is provided in email.
- (f) What are the goals of security?

Please Turn Over

Group - C

Answer *any four* questions.

3. (a) Write RSA algorithm with a suitable example.
(b) What is Message digest? Why is it used? 6+4
4. (a) Explain the Diffie-Hellman key exchange algorithm with an example.
(b) What is the purpose of x-boxes in DES? 6+4
5. Write short notes on (*any two*) : 5×2
(a) S/MIME
(b) CIA triad
(c) Differential cryptanalysis
(d) Public-Key Cryptosystem.
6. (a) What do you mean by Galois field?
(b) What is digital certificate? 8+2
7. (a) Explain Chinese-Remainder theorem.
(b) What are the limitations of firewall? 6+4
8. (a) What is masquerade? Which principle of security is breached because of that?
(b) Differentiate between SSL and SET.
(c) What are the two modes of IP Sec? Draw proper diagram to explain each mode. 2+4+4
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